Money Laundering Is Not an Outlier: Detecting Illicit Bitcoin Transactions Without Illicit Labels

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Abstract

Cryptocurrency exchanges must flag laundering transactions, but confirmed illicit labels are scarce and late, while licit ones are cheap. On the Elliptic Bitcoin graph (203,769 transactions, 49 time steps) we show that illicit transactions are *not* outliers: only 1.7% exceed a 3σ rule versus 22.3% of licit ones, so one-class detectors rank them *below* normal (test ROC-AUC 0.17). We therefore train a positive-unlabeled detector (LvU) that separates licit from unlabeled transactions and never sees an illicit label. On a chronological test split, LvU reaches PR-AUC 0.546 ± 0.024 (8.4× prevalence) and 82% precision in its top 100 alerts, statistically indistinguishable in PR-AUC from a supervised GraphSAGE network and recovering 69% of supervised XGBoost's PR-AUC. Ablations show that 1% illicit contamination of the licit labels cuts this PR-AUC by 74%, and every method fails after a dark-market shutdown.

1 Introduction

Scenario and stakeholders. Virtual-asset service providers must screen the transactions they touch for money laundering. Every alert goes to a compliance analyst, who decides whether to freeze funds and file a suspicious-activity report. A missed laundering transaction (FN) exposes the provider to regulatory penalties; a false alert (FP) costs analyst time from a fixed review budget. Licit labels come from the provider's own KYC-verified customers and from counterparties attributed to regulated exchanges; analyst capacity fixes the alert budget per period.

Motivation. Supervised detectors work well on Bitcoin data (Weber et al., 2019) but need confirmed illicit labels, which arrive late. One-class models, treating anomalies as rare far-away points (Chandola et al., 2009), are the label-free default, yet Lorenz et al. (2020) found them poor and illicit instances not outlying; we quantify, explain and exploit this.

Definition of anomaly. Our data belong to the **Graph** category: the unit of detection is one Bitcoin transaction, a node in a payment graph. It is anomalous if Elliptic attributes it to an illicit entity (scams, malware, ransomware, Ponzi schemes) and normal if to a licit one (exchanges, wallet providers, miners). Example normal sample: tx 378120377 (step 16, licit; `local_53`= 0.92, `local_55`= 0.45, in/out degree 0/2). Example anomaly: tx 232014511 (step 1, illicit; `local_53`= −0.49, `local_55`= −0.47, degree 1/1).

Research question. *Can a detector fitted without any illicit label rank illicit Bitcoin transactions above licit ones on future time steps, even though illicit transactions are not outliers?*

Contributions. (C1) Illicit transactions are less extreme than licit ones (1.7% vs. 22.3% beyond 3σ), which inverts outlier detectors (test ROC-AUC 0.17 for Isolation Forest, 0.33 for 3σ; §4, §7). (C2) A licit-vs-unlabeled (LvU) detector reaches test PR-AUC 0.546 and P@100 0.82 without illicit labels: far above label-free baselines, indistinguishable in PR-AUC from a supervised GNN, below supervised trees (§7). (C3) It is robust to the unlabeled pool size but collapses when 1% of licit labels are wrong (§7.1). (C4) 42% of its misses fall after a dark-market shutdown, a blind spot shared with supervised models (§8).

2 Related Work

Data category and literature. Our data are in the **Graph** category (transactions are nodes, payment flows directed edges); its key challenge is multi-hop structure, as laundering routes funds through chains of transactions. Weber et al. (2019) released Elliptic and compared supervised models on a 70:30 temporal split (Table 1); Random Forest

Work	Data	Paradigm	Method	F1
Weber et al. (2019)	Elliptic	sup.	RF, AF	0.788
Weber et al. (2019)	Elliptic	sup.	GCN	0.628
Weber et al. (2019)	Elliptic	sup.	EvolveGCN	0.720
Lorenz et al. (2020)	Elliptic	unsup.	LOF (best)	$\leq$0.19
Lorenz et al. (2020)	Elliptic	unsup.	IF	$\leq$0.01

Table 1: Prior results (illicit-class F1; sup./unsup. = (un)supervised). Prior work trains on steps 1–34; our split differs (Table 2), so results in Table 3 are not directly comparable.

Split	Steps	Tx	Licit	Illicit	Unknown	Ill./lab.
train	1–29	120,804	23,510	2,871	94,423	10.9%
val	30–34	15,461	2,922	591	11,948	16.8%
test	35–49	67,504	15,587	1,083	50,834	6.5%
all	1–49	203,769	42,019	4,545	157,205	9.8%

Table 2: Dataset summary (234,355 edges, 166 features) and the chronological split shared by all methods.

on all features beat graph convolutional networks. EvolveGCN (Pareja et al., 2020) models the graph sequence over time (illicit F1 0.720 in Weber et al., 2019) and reports that no model recovers after the dark-market shutdown. Lorenz et al. (2020) found seven unsupervised detectors at illicit F1 $\leq$0.19 and matched supervision with active learning on 5% of the labels. Positive-unlabeled (PU) learning trains a classifier from one labeled class plus unlabeled data (Elkan and Noto, 2008; Bekker and Davis, 2020); we do not use Deep SAD (Ruff et al., 2020), which trains on labeled anomalies that a provider without confirmed cases lacks.

Positioning and method selection. Tier 1 assumes anomalies are extreme in some feature (3σ) or nothing (random score). Tier 2 assumes illicit mass is over-represented in unlabeled relative to licit data (LvU); the one-class Isolation Forest (Liu et al., 2008) instead assumes anomalies are easy to isolate. Tier 3 is fully supervised: Random Forest (Breiman, 2001) as in Weber et al. (2019), XGBoost (Chen and Guestrin, 2016), and a GNN. Graph context enters as one-/two-hop aggregates and degree (A4) and through GCN (Kipf and Welling, 2017) or GraphSAGE (Hamilton et al., 2017) message passing, both supervised (T3c) and inside LvU (T2″). As unknown nodes dominate neighbourhoods (§4), we expect a node-level PU classifier to suit this graph. Unlike prior work, we fit without illicit labels, explain *why* outlier detectors invert, and analyse failures per transaction.

3 Task Definition and Data

Data set. We use the Elliptic Data Set (Weber et al., 2019; Elliptic, 2019) (CC BY-NC-ND 4.0; an unmodified copy is in our repository). It contains $N = 203{,}769$ transactions and 234,355 edges in 49 time steps about two weeks apart; no edge crosses time steps (verified: 0 of 234,355), so graph features never mix periods. Each node has $D = 166$ anonymised features: the time step, 93 local features (LF; e.g. fees, input/output counts) and 72 one-hop neighbour aggregates (AF); we add in/out degree and two-hop aggregates (mean of the neighbours' AF). Unit and examples: §1.

Labels. Heuristic entity-level labelling (Weber et al., 2019) gives 4,545 illicit, 42,019 licit and 157,205 *unknown* transactions; labels may be noisy, and unknown ones (both classes) are never evaluated but serve LvU as unlabeled data. The paradigm is semi-supervised (PU): licit and unlabeled data for fitting, illicit labels only for validation-based selection.

Class imbalance. Overall, 2.2% of transactions are illicit; among labeled ones the ratio is 9.8% vs. 90.2% (1 : 9.2), and on the test split 6.5% (1 : 14.4). "Always licit" reaches 93.5% test accuracy with zero recall, so we report PR-AUC and tuned-τ metrics, never accuracy or a 0.5 cut-off; as licit rows dominate an unweighted log-loss, every τ is tuned on validation (GNNs use a class-balanced loss).

4 Exploratory Feature Analysis

All statistics use the training split only (Figure 1).

Distribution contrast. Figure 1a shows `local_53`, the feature with the largest Kolmogorov–Smirnov (KS) distance between classes (0.66, mutual information 0.21 nats). Most illicit transactions sit on one value (-0.49 ± 0.001: 77% of illicit, 16% of licit training transactions); licit values spread widely. Figure 1b is the key observation: only 1.7% of illicit transactions have any local feature beyond 3σ of the licit distribution, against 22.3% of licit transactions. In PCA space (Figure 1c), illicit points form one compact region.

Separability and redundancy. Local features separate the classes better than one-hop aggregates (mean KS 0.38 vs. 0.27; 63 vs. 30 features with KS > 0.3; mean MI 0.088 vs. 0.060 nats) and than degree or two-hop aggregates (mean KS 0.20, 0.18). 89 feature pairs have $|r| > 0.95$, and licit

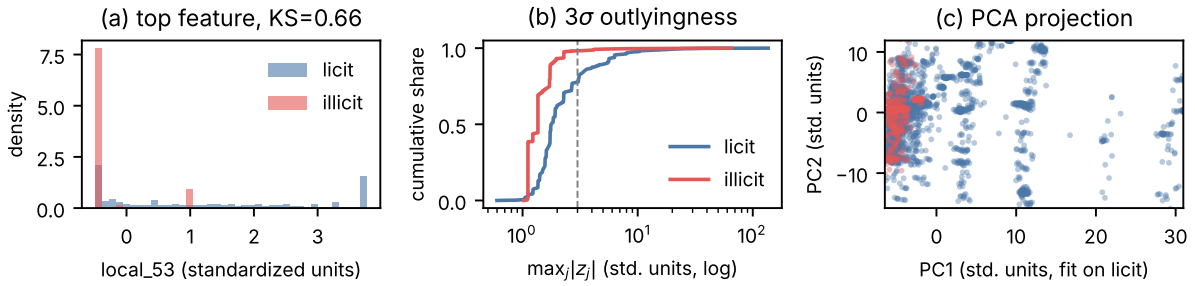


Figure 1: Training split only. (a) The most separating feature: illicit transactions concentrate on one value. (b) Cumulative share of the largest $|z|$-score (w.r.t. licit statistics): illicit transactions are *less* extreme than licit ones; dashed line = 3σ. (c) PCA fitted on licit data: illicit points form a compact cluster inside the region occupied by licit ones.

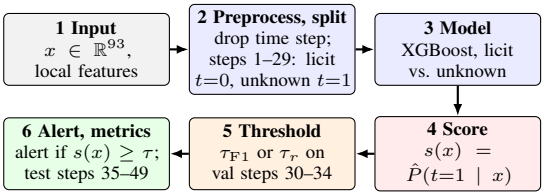


Figure 2: LvU pipeline. No illicit label is used for fitting; labeled validation transactions only select hyperparameters and set τ.

behaviour drifts moderately within training (mean KS 0.11, steps 1–14 vs. 15–29). On training edges, 56% of the labeled neighbours of illicit transactions are illicit (5× the base rate), but 59% of all their neighbours are unknown (licit: 46%).

Hypotheses. **H1**: Outlier-based scores (3σ, Isolation Forest, Mahalanobis) will rank illicit transactions *below* licit ones. **H2**: A model that contrasts licit data with unlabeled data will rank them above, because the compact illicit mode is denser in the unlabeled pool than among labeled licit transactions. **H3**: Graph context (aggregates, degree, message passing) will add little, as unknown neighbours dominate.

5 Method

Pipeline. Figure 2 shows the six checkpoints of the Tier 2 pipeline. The input is the 93 local features as released (provider-scaled per feature, to which tree rankings are invariant); the time step is dropped so the model cannot memorise periods. In the training steps, labeled licit transactions get target $t = 0$ and unknown ones $t = 1$ (23,510 vs. 94,423 rows); illicit training rows are *not* used. An XGBoost classifier (300 trees, depth 6, learning rate 0.1, row/column subsampling 0.8) learns $\hat{P}(t{=}1 \mid x)$.

Anomaly score. For a well-calibrated classifier, $s(x) = \hat{P}(t{=}1 \mid x)$ is monotone in the density ratio $p_{\mathcal{U}}(x)/p_{\mathcal{L}}(x)$. If the unknown pool is a mixture $p_{\mathcal{U}} = \pi\, p_{\text{ill}} + (1-\pi)\, p_{\text{lic}}$ and labeled licit data follow p_{lic}, the ratio equals $(1-\pi) + \pi\, p_{\text{ill}}(x)/p_{\text{lic}}(x)$ and ranks transactions by how much likelier they are under the illicit than the licit distribution (Elkan and Noto, 2008), rewarding a dense illicit mode that licit data rarely reach (H2).

Threshold and error costs. An FN lets laundered funds pass (regulatory risk); an FP costs analyst time. With cost ratio $r = c_{\text{FN}}/c_{\text{FP}}$, the cost-optimal τ_r minimises $r \cdot \text{FN} + \text{FP}$ on the labeled validation transactions (steps 30–34; the test split is never used). Tiers are compared at the validation-F1 maximiser τ_{F1} (0.958 ± 0.012 for LvU), which weighs both errors about equally ($r \approx 1$); deployment should use τ_{10}, which suppresses the costlier FN (§7.1).

6 Experimental Setup

Split and leakage prevention. All methods share one chronological split by time step (Table 2): train 1–29, validation 30–34, test 35–49, i.e. 59/10/31% of steps and 57/8/36% of labeled transactions. Training rows per tier (licit / illicit / unknown, steps 1–29): 3σ, one-class IF, Mahalanobis and LOF 23,510 / 0 / 0 (normal-only); LvU and LvU-GraphSAGE 23,510 / 0 / 94,423 (unknown may hide illicit); RF, XGBoost and supervised GraphSAGE 23,510 / 2,871 / 0. Every tier tunes on the same 2,922 / 591 validation and is scored on the same 15,587 / 1,083 test transactions; only A1 mixes illicit rows into the licit side. Detector scalers use licit training rows; the test set is scored once per seed.

Configurations. LvU: feature set $\in$\{LF, AF, AF+degree\}, depth $\in \{4, 6, 8\}$ and trees $\in \{100, 300\}$, chosen by validation PR-AUC (LF, 6, 300; seed-0 search 0.901, 5-seed mean 0.871). Isolation Forest: 200 trees, `max_samples` $\in \{256, 2048\}$ (2048, LF). RF as in Weber et al. (2019) (50 trees, 50 features per split, AF); XGBoost as LvU, on AF. GNNs (PyTorch Geometric, Fey and Lenssen, 2019): two GCN or GraphSAGE layers, full-batch, Adam (rate 0.01), early stopping on validation PR-AUC ($\leq$1000 epochs, patience 100); validation picked GraphSAGE, width 64 and AF (the supervised grid also tried width 128). Depth 4–8 and 100–300 trees span shallow to moderate boosting for $\sim 10^5$ rows; Tier 3 reuses Weber's RF and LvU's XGBoost settings (on AF; an LF control is in §7). Stochastic methods use 5 seeds; deterministic ones (3σ, Mahalanobis, LOF, kNN ratio) report a 1,000-resample test bootstrap std (†). Software: Python 3.13, scikit-learn 1.9 (Pedregosa et al., 2011), XGBoost 3.4, PyTorch 2.14; hardware: 4 vCPUs, 16 GB RAM, no GPU.

Metrics. **PR-AUC** (primary) ignores the many easy true negatives and suits imbalance (Saito and Rehmsmeier, 2015); its random baseline is the prevalence. **P@100**, the precision of the 100 top-scored test transactions, models a fixed analyst budget; **F1@τ** evaluates the deployed decision; ROC-AUC shows inversion (< 0.5).

7 Results

Main comparison. Table 3 supports H1: 3σ and the one-class Isolation Forest rank illicit below licit transactions (ROC-AUC 0.33, 0.17) and fall under random PR-AUC. The 3σ rule (fixed $\tau = 3$) reaches F1 0.010, and Isolation Forest's val-tuned τ degenerates to "alert on everything" (F1 0.121, as for random scores). LvU supports H2: PR-AUC rises to 0.546 ± 0.024 (8.4$\times$ prevalence) and 82% of its top-100 alerts are illicit. Bootstrap 95% CIs of the PR-AUC gap from LvU: random [0.45, 0.51], 3σ [0.46, 0.53], IF [0.48, 0.54], RF [-0.24, -0.20], XGBoost [-0.27, -0.22], GraphSAGE [-0.04, 0.03]; paired t-tests over seeds agree ($p < 10^{-4}$ for every stochastic method except GraphSAGE).

Against Tier 3. Supervised trees remain clearly better (XGBoost 0.790, RF 0.767), though LvU recovers 69% of XGBoost's PR-AUC without illicit labels. The gap is not a feature confound: on LF, as LvU uses, RF and XGBoost still reach 0.742 and 0.766 (5 seeds, std $\leq$0.003). It loses because it learns "unlabeled-like", not "illicit" (verified in §8). RF's F1 varies across seeds (0.42–0.80) unlike its PR-AUC: a τ tuned at 16.8% validation prevalence transfers poorly to 6.5%. Trees train in $<$6 s, GNNs in 2–5 min; all score in $\leq$12 μs per transaction.

Message passing (Graph-specific). Supervised GraphSAGE (T3c, PR-AUC 0.552 ± 0.035) is indistinguishable from label-free LvU ($p = 0.79$; CI [-0.038, 0.027]) and 0.24 below XGBoost, as trees beat GNNs on Elliptic (Weber et al., 2019). Inside LvU, message passing fails (T2″, 0.189): averaging unknown-dominated neighbourhoods makes most nodes look "unlabeled" (H3).

7.1 Controlled Ablations

Table 4 varies one factor at a time; the configuration was fixed on validation (§6) before any ablation, test labels never chose among settings, and every conclusion also holds on validation (Val).

A1: contamination of the normal labels (AD-specific). Swapping 1% of licit training rows for illicit ones (235 rows) cuts PR-AUC from 0.546 to 0.142. As illicit transactions form one compact mode, a few hundred of them labeled $t = 0$ teach the classifier that this mode is "licit": LvU needs clean licit labels.

A2: size of the unlabeled pool (AD-specific). 5% of the unknown pool (4.7k rows) gives the same PR-AUC (0.546); no change is significant (Welch t-test over seeds, $p \geq 0.36$), so a small unlabeled sample already holds the illicit mode.

A3: scoring function (AD-specific). With features fixed (one-class on licit rows, PU on licit+unlabeled), outlier scores invert (Isolation Forest 0.037; Mahalanobis with Ledoit–Wolf shrinkage, Ledoit and Wolf, 2004, 0.038), local density scores (LOF with k=20, Breunig et al., 2000; a licit/unlabeled 10-NN density ratio) recover some signal (0.19), and the classifier-based density ratio is best (0.546).

A4: graph features (Graph-specific). Relative to LF, AF ($+0.011$), AF+degree (-0.037) and AF+two-hop (-0.010) change test PR-AUC insignificantly (Welch $p \geq 0.18$) and lower test F1 (0.30–0.35 vs. 0.40), which supports H3; on validation, LF is clearly best (0.871 vs. $\leq$0.783), which is why it was selected.

Tier	Method	PR-AUC	ROC-AUC	F1@τ	P@100	PR-AUC, steps 35–42	PR-AUC, steps 43–49	F1, steps 35–42	F1, steps 43–49	Fit (s)	Score (μs/tx)
T1a	Random score	0.065±0.001	0.495±0.006	0.122±0.000	0.060±0.017	0.091	0.027	0.17	0.05	0.1	0.0
T1b	3σ rule	0.053±0.002†	0.326±0.008†	0.010±0.002†	0.040±0.020†	0.075	0.028	0.01	0.01	0.1	0.2
T2′	One-class IF	0.038±0.000	0.170±0.008	0.121±0.002	0.000	0.053	0.021	0.17	0.05	0.7	11.9
T2″	LvU-GraphSAGE	0.189±0.017	0.799±0.018	0.279±0.023	0.190±0.046	0.289	0.036	0.39	0.03	120	1.6
T2	**LvU (ours)**	0.546±0.024	0.865±0.002	0.398±0.011	0.822±0.020	0.743	0.025	0.54	0.01	4.6	1.5
T3a	RF (sup.)	0.767±0.003	0.891±0.004	0.541±0.138	**1.000**	0.897	0.028	0.70	0.02	5.7	2.2
T3b	XGBoost (sup.)	**0.790±0.001**	**0.922±0.002**	**0.675±0.056**	**1.000**	**0.914**	0.036	**0.81**	0.03	2.8	1.7
T3c	GraphSAGE (sup.)	0.552±0.035	0.874±0.003	0.568±0.028	0.842±0.086	0.687	0.029	0.65	0.01	291	1.6
prevalence		0.065				0.092	0.025				

Table 3: Test results (labeled transactions of steps 35–49), mean ± std over 5 seeds; †: std over 1,000 test bootstrap resamples for the deterministic 3σ rule (all metrics). F1 uses τ_{F1} tuned on validation (3σ rule: $\tau = 3$). Best per quality metric in bold (not for steps 43–49, where all methods are near chance); prevalence = random PR-AUC. Period columns split the test set at the dark-market shutdown (step 43); period F1 pools alerts within each period.

Setting	Test PR-AUC	Δ	Val
A1: illicit rows hidden in licit side			
0% (ours)	0.546±0.024	—	0.871
1%	0.142±0.007	−0.404	0.359
5%	0.098±0.003	−0.448	0.247
10%	0.091±0.004	−0.455	0.217
A2: fraction of unlabeled pool used			
5%	0.546±0.018	+0.000	0.849
25%	0.520±0.047	−0.026	0.863
50%	0.551±0.040	+0.005	0.833
100% (ours)	0.546±0.024	—	0.871
A3: label-free scoring function			
one-class IF	0.037±0.000	−0.509	0.108
one-class Mahalanobis	0.038±0.001†	−0.508	0.097
one-class LOF	0.190±0.010†	−0.356	0.218
PU kNN ratio	0.194±0.010†	−0.352	0.713
PU classifier (ours)	0.546±0.024	—	0.871
A4: feature set			
LF, 93 local (ours)	0.546±0.024	—	0.871
AF (+72 aggregates)	0.557±0.037	+0.011	0.768
AF + in/out degree	0.509±0.043	−0.037	0.756
AF + 2-hop aggregates	0.536±0.046	−0.010	0.783

Table 4: Ablations of LvU, one factor at a time: test PR-AUC (5 seeds; † bootstrap std), Δ vs. the full method, and validation PR-AUC (Val, mean). A3's IF uses the default `max_samples`=256.

A5: threshold rule. In Table 6, τ_1's F1 does not differ significantly from τ_{F1}'s ($p = 0.22$), so τ_{F1} acts like $r \approx 1$; every other rule shifts F1 ($p < 0.001$), trading false alerts for recall. For a compliance provider we recommend fixing r explicitly; with $r = 10$ (a miss costs ten reviews), τ_{10} raises 3,952 alerts (≈263 per two-week step) for recall 0.75 at 3.9 false alerts per catch.

8 Analysis

Temporal failure. As in Weber et al. (2019), every model fails after a dark market closes at step 43 (Table 3): LvU's F1 drops from 0.54 to 0.01 and its PR-AUC from 0.743 to 0.025, the prevalence. The new illicit population leaves the illicit mode: `local_53`$= -0.49$ holds for 84% of illicit test transactions before step 43 but only 2% after (training: 77%), so for the 166 post-shutdown misses this feature contributes −0.33 log-odds instead of +1.03 for earlier hits. Steps 43–49 hold 15.6% of test illicit transactions but 42% of the 398 misses (seed 0); 69% of misses are shared by supervised RF, so the blind spot lies in the data.

Failure cases. In Table 5, #1 is a complete inversion ($s - \tau = -0.956$): `local_48` (TreeSHAP −1.54 log-odds) and `local_53`$= 1.81$ (−1.53), above 99.7% of illicit but at the 63rd percentile of licit training transactions, pull its score down; RF misses it too (0.04). #2 (FN) and #4 (FP) fall on either side of $\tau = 0.97027$ ($s - \tau = -5.1\times10^{-4}$ and $+3.8\times10^{-5}$): they share 73 of 93 local features (within 10^{-4}; L2 distance 4.1 vs. a median of 10.2 to licit training transactions) yet carry opposite labels. Label noise or shared wallet software may explain this, unverifiably (entities are anonymised). #3 is a confident FP ($s = 0.99995$) on the illicit mode `local_53`$= -0.49$ that RF scores 0.03. The "unlabeled-like" mechanism shows directly: among their 10 nearest training neighbours, licit FPs have 94% unknown transactions, licit TNs 47% (pool: 80%). **Shared pattern**: errors stem from overlapping modes and post-shutdown drift, not capacity; the first fix is a few analyst-confirmed labels from the new period (active learning, Lorenz et al., 2020).

Interpretability. By exact TreeSHAP of Lundberg et al. (2020) (mean |SHAP| over test illicit transactions), `local_53` (0.88 log-odds),

#	Type	txId	Step	$s(x)$	$s-\tau$	Top SHAP	RF
1	FN	97069571	40	0.01426	-0.956	L48= 0.09 (-1.54)	0.04
2	FN	73003591	42	0.96975	-5.1×10^{-4}	L53= -0.49 ($+0.87$)	0.76
3	FP	169360875	45	0.99995	$+0.030$	L53= -0.49 ($+1.23$)	0.03
4	FP	73160604	42	0.97030	$+3.8\times10^{-5}$	L53= -0.49 ($+0.85$)	0.41

Table 5: Failure cases of LvU (seed 0, $\tau = 0.97027$; alert if $s \geq \tau$); the text refers to them by key #1–#4. RF = supervised Random Forest score (mean of seeds; mean $\tau \approx 0.37$). Top SHAP = local feature Lk (`local_`k) with the largest |TreeSHAP|: value (attribution in log-odds).

Alert rule	Alerts	Prec.	Rec.	F1	FP/catch	p
τ_{F1}	2,459	0.29	0.65	0.40	2.5	—
τ_1	2,333	0.30	0.64	0.41	2.4	0.22
τ_5	3,149	0.24	0.71	0.36	3.1	$<$0.001
τ_{10}	3,952	0.20	0.75	0.32	3.9	$<$0.001
τ_{20}	4,979	0.17	0.78	0.28	4.8	$<$0.001
licit 95th pct.	5,420	0.16	0.80	0.27	5.3	$<$0.001
top 30/step	450	0.48	0.20	0.28	1.1	$<$0.001

Table 6: Alert rules for LvU on the test split (15 steps), mean of 5 seeds (precision/recall std $\leq$0.04); τ_r minimises $r \cdot \text{FN} + \text{FP}$ on validation; p: paired t-test of F1 vs. τ_{F1}.

`local_47` and `local_76` (0.56) dominate: the model uses the illicit mode of §4. For the top true positive (tx 162428632, $s = 0.99990$, RF 0.94), `local_53`$= -0.49$ adds 1.23 log-odds, five others 0.4–0.9. An analyst escalates when the attribution sits on the known illicit mode and dismisses when a licit counterparty explains it: a reason to review, not proof.

Practical impact. At τ_{F1}, LvU raises 2,459 alerts in 15 steps ($\approx$164 per step) and catches 65% ($\approx$704) of 1,083 illicit test transactions; 71% of alerts are false (2.5 per catch). With 30 alerts per step, LvU catches 218 illicit transactions (precision 0.48, 7.4$\times$ random review), supervised RF 239 and XGBoost 245: lacking illicit labels costs 21–27 catches per 450 reviews, so LvU suits cold-start triage.

9 Conclusion and Limitations

Key findings. Yes, with qualifications: fitted only on licit and unlabeled transactions, LvU ranks future illicit transactions far above chance (PR-AUC 0.546 vs. 0.065; P@100 0.82) where outlier detectors invert, because illicit transactions form a compact, non-extreme mode. It is indistinguishable in PR-AUC from a supervised GraphSAGE and reaches 69% of XGBoost's, but fails after the dark-market shutdown like every model. Verdicts: H1 and H2 supported; H3 holds for LvU but not for supervised trees (one-hop aggregates add 0.024–0.026).

Reflection and future work. The decisive step was testing outlyingness in EDA (Figure 1b), which we would now do first. Next: (i) after a drift alarm, query analyst labels for top-scored unlabeled transactions to fix post-shutdown misses; (ii) estimate the unlabeled class prior π (Bekker and Davis, 2020) to calibrate τ across periods.

Limitations. *Data coverage*: one blockchain, 49 steps, heuristic labels, anonymised features (with real ones we would add address-cluster age and velocity); 77% of transactions are unlabeled, so scores refer to labeled ones only. *Assumptions*: LvU assumes clean licit labels (A1 shows 1% noise is fatal) and that the unlabeled pool contains the illicit modes of the future, which fails after the shutdown. *Graph*: deeper or temporal GNNs (e.g. EvolveGCN) are untested. *Generalisation*: extreme fraud (e.g. one-off thefts) may not invert.

Deployment considerations. *Concept drift*: behaviour changed abruptly at step 43; audit the top 30 alerts per step and retrain when their precision halves, as scores alone hide a collapse. *Inputs*: the unlabeled pool is every other on-chain transaction; 30 alerts per step is $\approx$3 reviews per working day. *Latency, privacy*: $\approx$1.5 μs per transaction; linking addresses to people is regulated.

Contributions of Each Member

- Member A: EDA; vetted the topic.
- Member B: assessed candidate topics.
- Member C: presentation slides.
- Member D: drafted the paper.
- Member E: code, slides, coordination.
- Member F: coordination, review.
- Member G: review, topic vetting.
- Member H: obtained the dataset.

AI usage statement. We used Claude Code for code and drafting; the team chose the question and verified every number and citation.

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